Chapter 18, Problem 23P

Problem

If the Fourier transform of $f(t)$ is $F(\omega) = \dfrac{10}{(2+j\omega)(5+j\omega)}$ determine the transforms of the following:

(a) $f(-3t)$

(b) $f(2t-1)$

(c) $f(t)\cos 2t$

(d) $\dfrac{d}{dt}f(t)$

(e) $\displaystyle\int_{-\infty}^{t} f(t)\,dt$

Step-by-step solution

Step 1 of 5

(A) $f(3t)\,leads\ to\ \dfrac{1}{3}\cdot\dfrac{10}{\left(2+\dfrac{j\omega}{3}\right)\left(5+\dfrac{j\omega}{3}\right)}-\dfrac{30}{(6+j\omega)(15+j\omega)}$

$$F\left[f(-3t)\right]=\frac{30}{(6-j\omega)(15-j\omega)}$$

Step 2 of 5

(B) $f(2t)\rightarrow\dfrac{1}{2}\cdot\dfrac{10}{\left(2+\dfrac{j\omega}{2}\right)\left(5+\dfrac{j\omega}{2}\right)}=\dfrac{20}{(4+j\omega)(10+j\omega)}$

$$f(2t-1)=f\left[2\left(t-\frac{1}{2}\right)\right]\rightarrow\frac{20e^{-\frac{j\omega}{2}}}{(4+j\omega)(10+j\omega)}$$

Step 3 of 5

(C) $=f(t)\cos 2t\rightarrow\dfrac{1}{2}F(\omega-2)+\dfrac{1}{2}F(\omega+2)$

$$=\frac{5}{\left[2+j(\omega+2)\right]\left[5+j(\omega-2)\right]}+\frac{5}{\left[2+j(\omega-2)\right]\left[5+j(\omega-2)\right]}$$

Step 4 of 5

(D) $F\left[f'(t)\right]=j\omega F(\omega)=\dfrac{j\omega 10}{(2+j\omega)(5+j\omega)}$

(E) $= \int_{-\infty}^{t} f(t)f(t)\,dt \to \dfrac{F(\omega)}{j\omega} + \pi F(0)\delta(\omega)$

$$= \frac{10}{j\omega(2+j\omega)(5+j\omega)} + \pi\delta(\omega) \times \frac{10}{2\times 5}$$

$$= \frac{10}{j\omega(2+j\omega)(5+j\omega)} + \pi\delta(\omega)$$